DO NOW — RECALL

Lesson 081 · Merge Sort: Bottom-up Merge Tracing

Name: _____ Class: _____ Date: _____

1. What is a sorted run?

2. How does insertion sort work?

3. What is O(n log n) complexity?

4. Binary search sorted [1,3,5,7,9,11,13] for 11: show midpoint index sequence.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Contiguous segment already in ascending order.

2. Model answer

Inserts each next key into its sorted prefix.

3. Model answer

Work scaling with n times log n.

4. Model answer

mid3=7 then mid5=11; return index5.

By the end, you can…

• Merge sort divides the input into smaller runs; merging two already sorted runs repeatedly selects the smallest available front value.

• Bottom-up merge sort begins with runs of length one and doubles run width on every complete pass; it does not require recursive calls.

• Worst-case time O(n log n); standard array merging uses O(n) extra space. Include comparisons, not just the final sorted sequence.

Vocabulary — know these terms

merge sort — Sort by merging sorted runs.

run — Consecutive already sorted segment.

merge — Combine two sorted sequences.

auxiliary space — Extra memory beyond input.

NEW LEARNING

Read, annotate and explain

Observe

Merge sort divides the input into smaller runs; merging two already sorted runs repeatedly selects the smallest available front value.

Apply

This bottom-up variant starts with single-item runs and doubles run width; it needs no recursive calls. The top-down variant recursively splits to length 0/1, then merges on return; Lesson 186 connects the two.

Evaluate

Worst-case time O(n log n); standard array merging uses O(n) extra space. Include comparisons, not just the final sorted sequence.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Bottom-up merge [8,3,6,1]; show pair results and full result.

Method and result

Pairs [3,8] and [1,6] → [3,8,1,6]; merge fronts → [1,3,6,8].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Given [7,2,9,1], state array after width1 pair merges and final width2 merge. [6]

Question 2

Merge sorted runs [2,7] and [1,9] and list chosen values in order, with comparisons. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare bottom-up merge with insertion sort for large random data. Give worst time and auxiliary-space cost. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Merge sort requires no additional array storage in its standard array implementation.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.